Levante

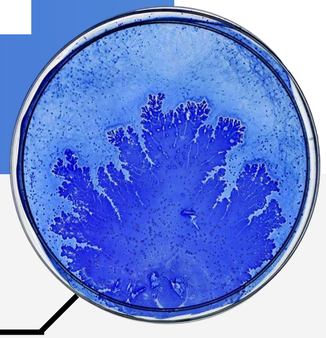
Edge AI Rapid Bacteria Detection for Beach Seawater



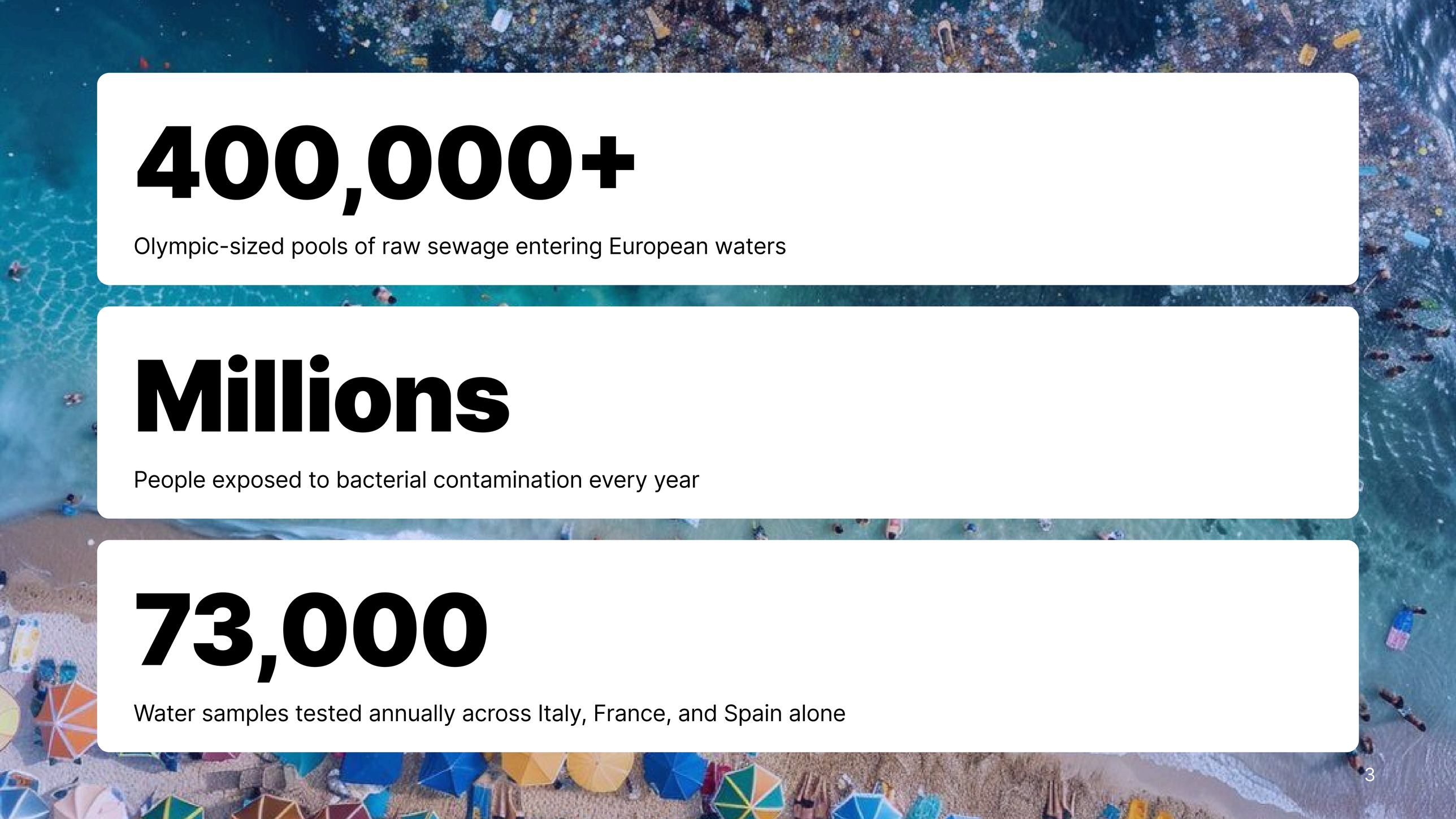
Group3

Exposed

24-48H



Confirmed



400,000+

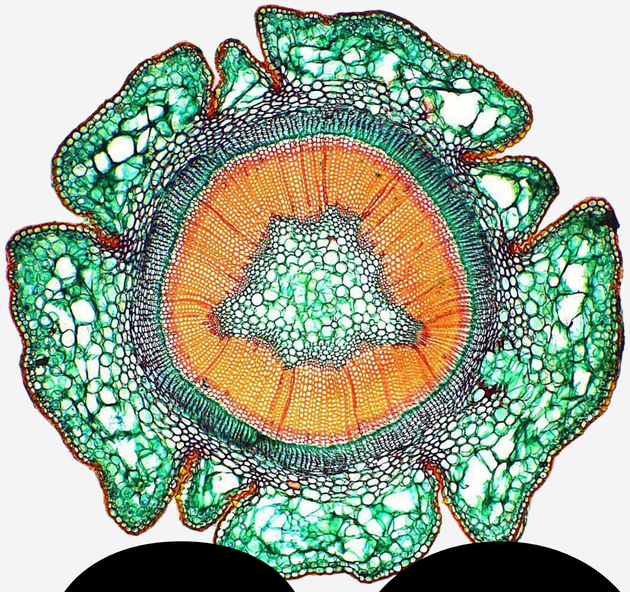
Olympic-sized pools of raw sewage entering European waters

Millions

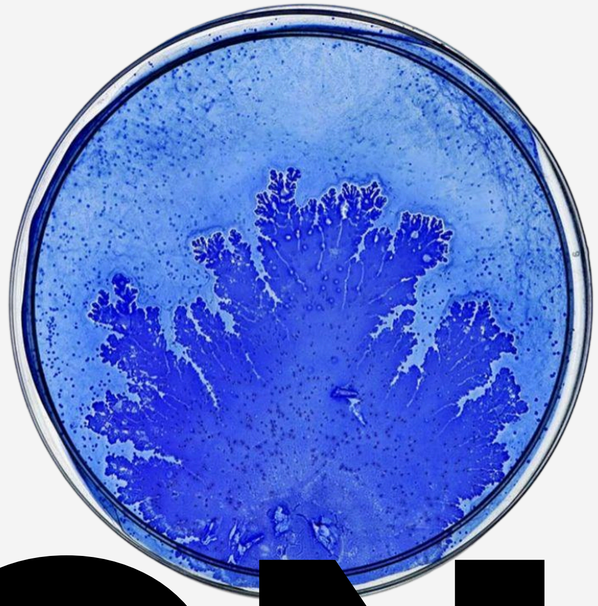
People exposed to bacterial contamination every year

73,000

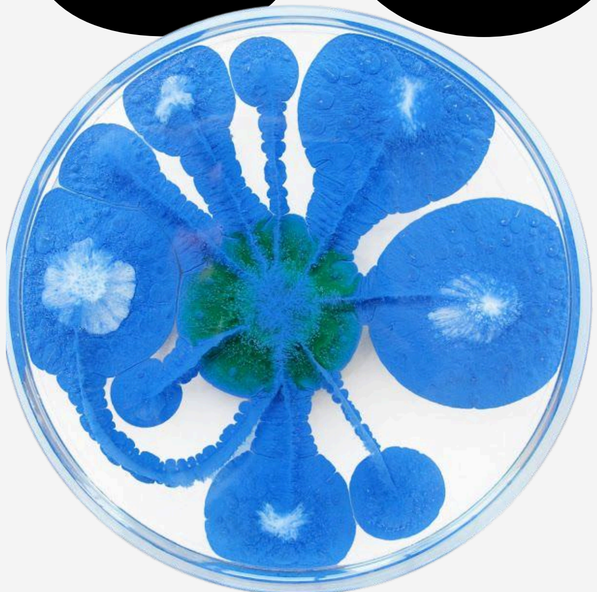
Water samples tested annually across Italy, France, and Spain alone

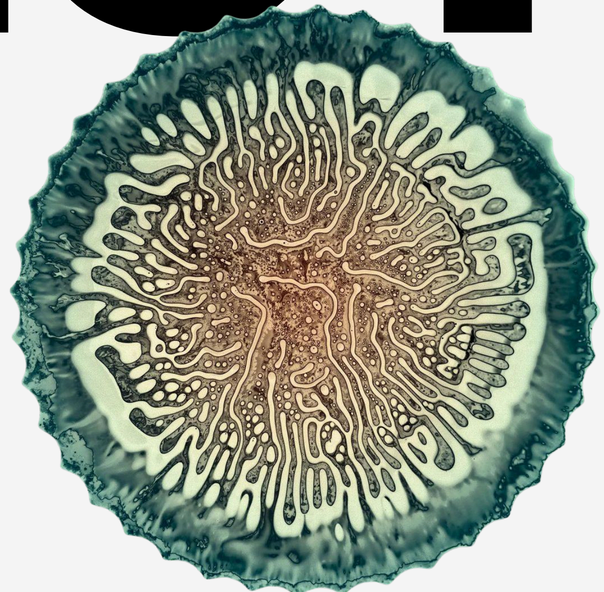






SOLUTION







“My main concern is my child’s health .”

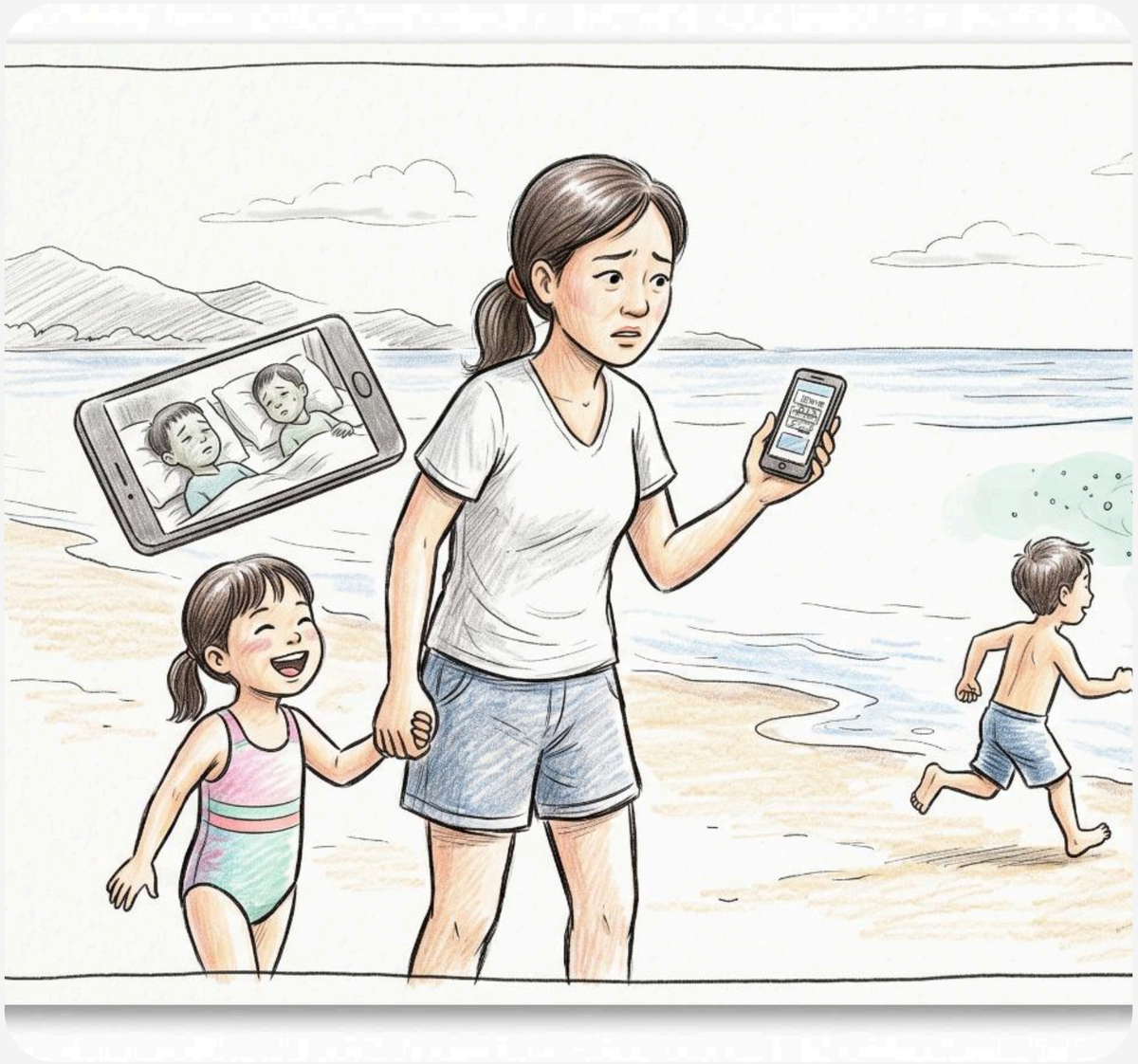
I need to know if the beach is safe as soon as possible.

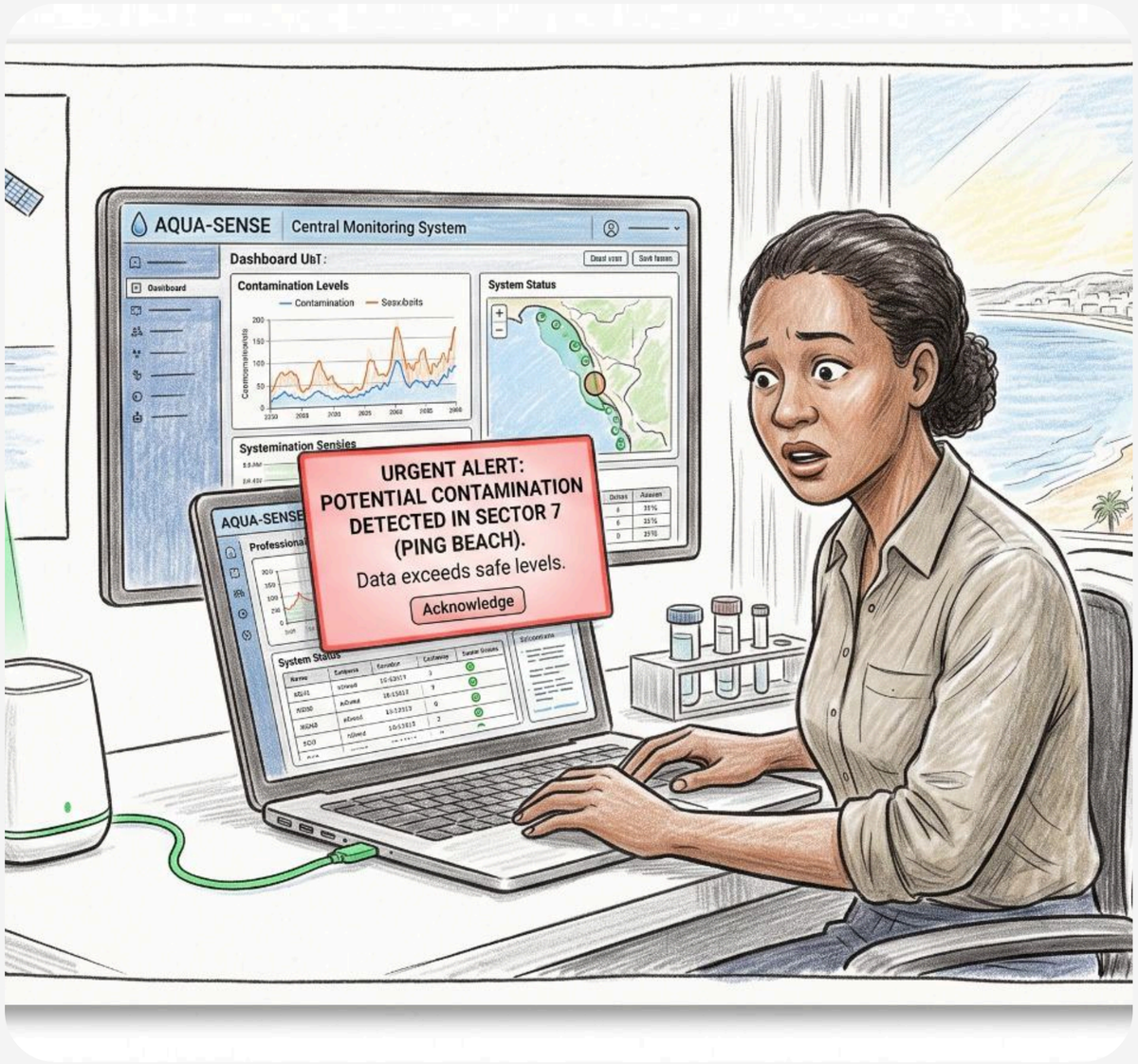




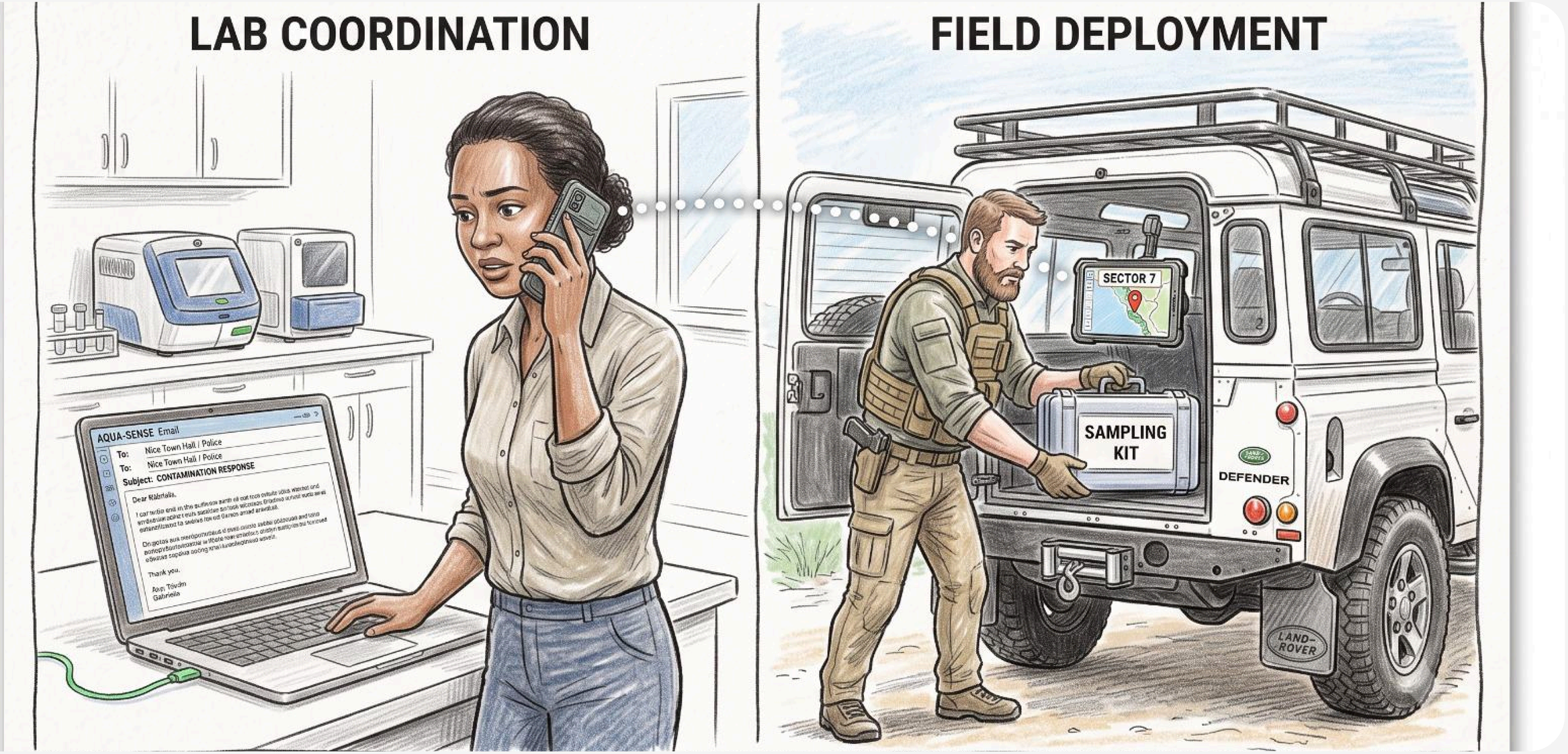
“I want to work more efficiently.”

Sampling and analysis take too long, plus experiments are costly.

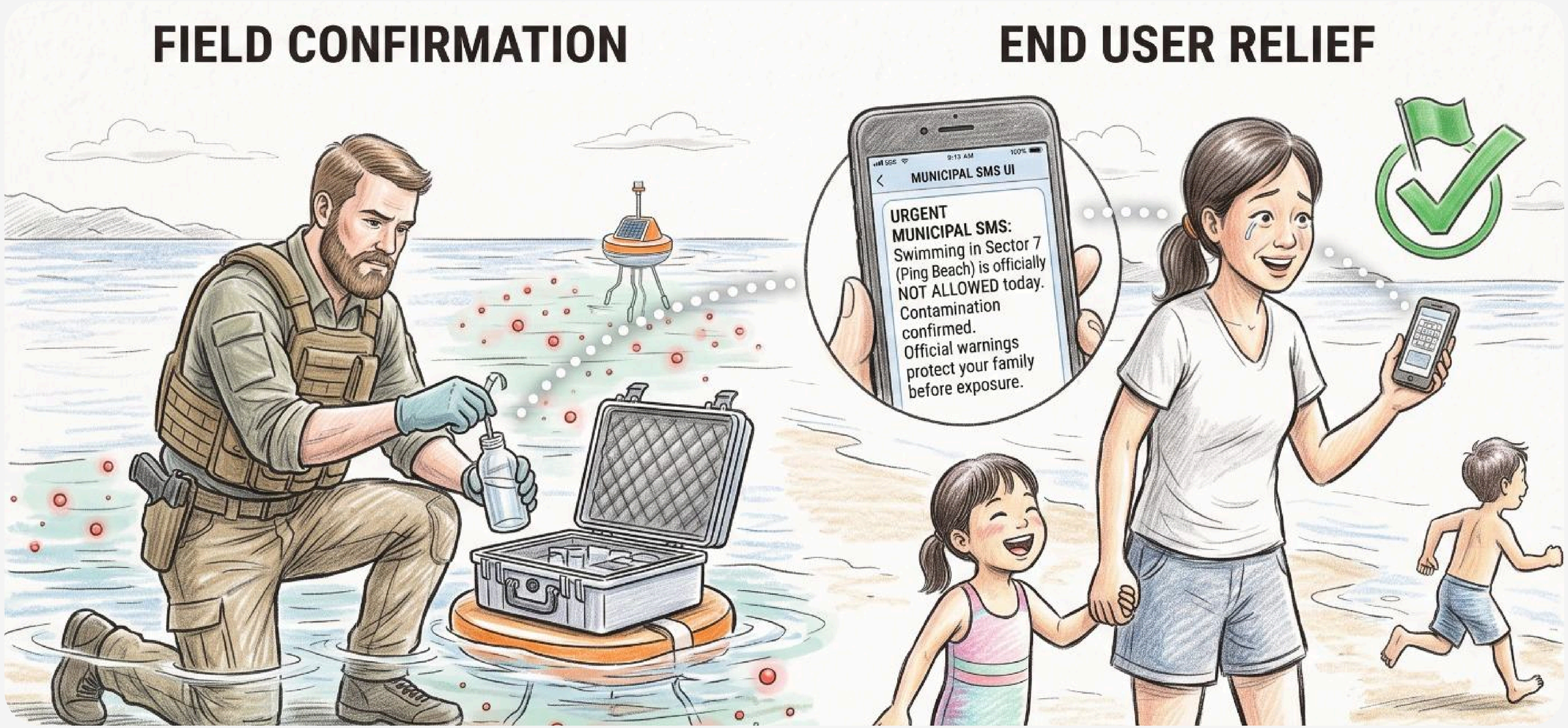



AQUA-SENSE
Central Monitoring System
Dashboard
Contamination Levels
System Status
URGENT ALERT:
POTENTIAL CONTAMINATION
DETECTED IN SECTOR 7
(PING BEACH).
Data exceeds safe levels.
Acknowledge

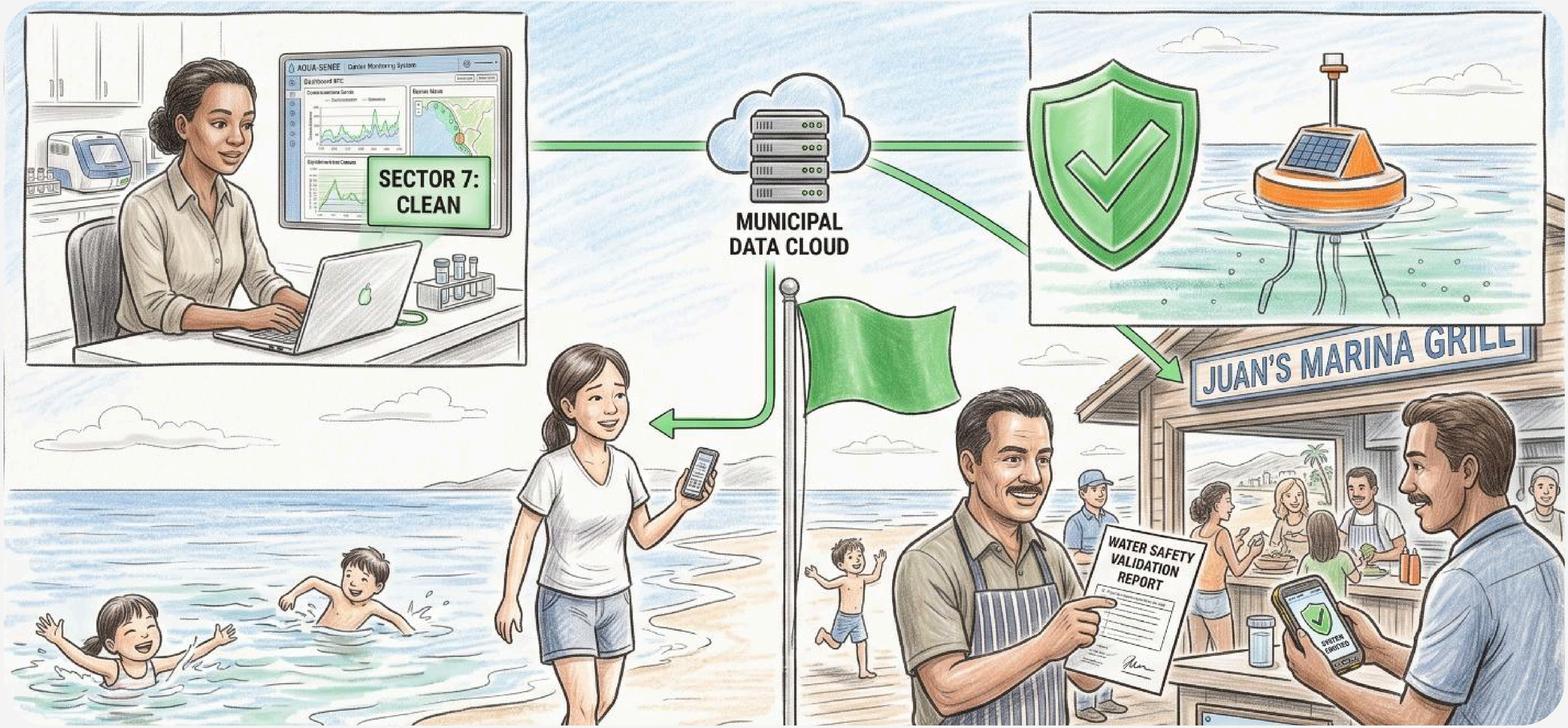
STORYBOARD


LAB COORDINATION
FIELD DEPLOYMENT
AQUA-SENSE Email
To: Nice Town Hall / Police
To: Nice Town Hall / Police
Subject: CONTAMINATION RESPONSE
Dear Mairie,
Thank you.
SECTOR 7
SAMPLING KIT
DEFENDER
LAND-ROVER

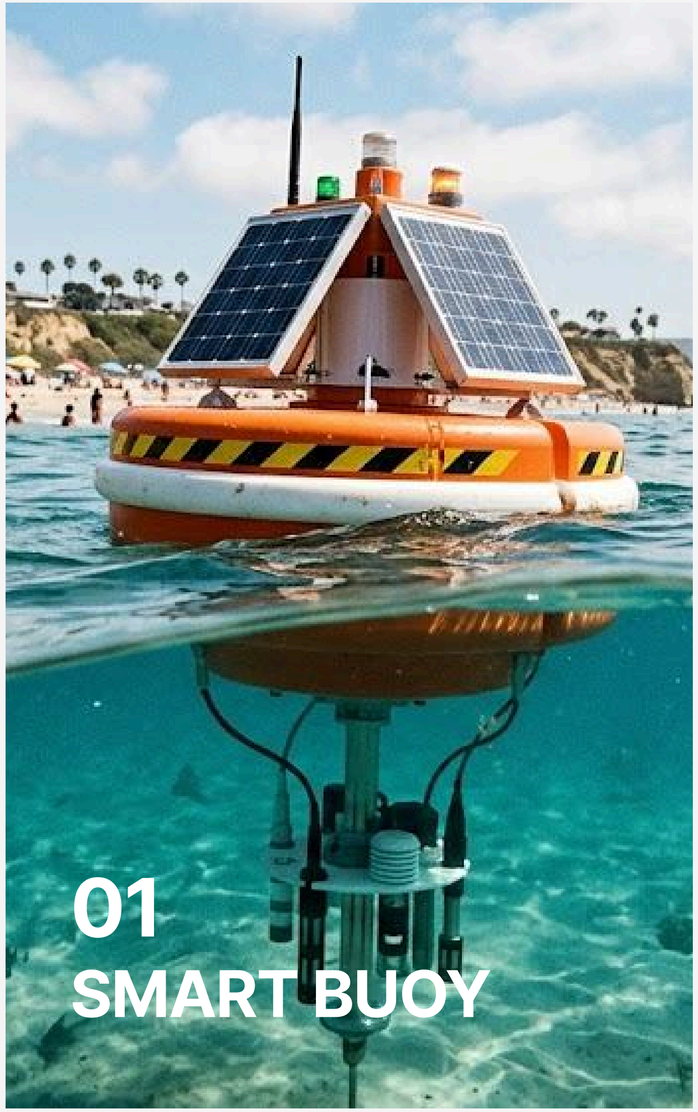
STORYBOARD


FIELD CONFIRMATION
END USER RELIEF
MUNICIPAL SMS UI
URGENT
MUNICIPAL SMS:
Swimming in Sector 7 (Ping Beach) is officially NOT ALLOWED today. Contamination confirmed. Official warnings protect your family before exposure.

STORYBOARD


AQUA-SENEE
SECTOR 7: CLEAN
MUNICIPAL DATA CLOUD
JUAN'S MARINA GRILL
WATER SAFETY VALIDATION REPORT

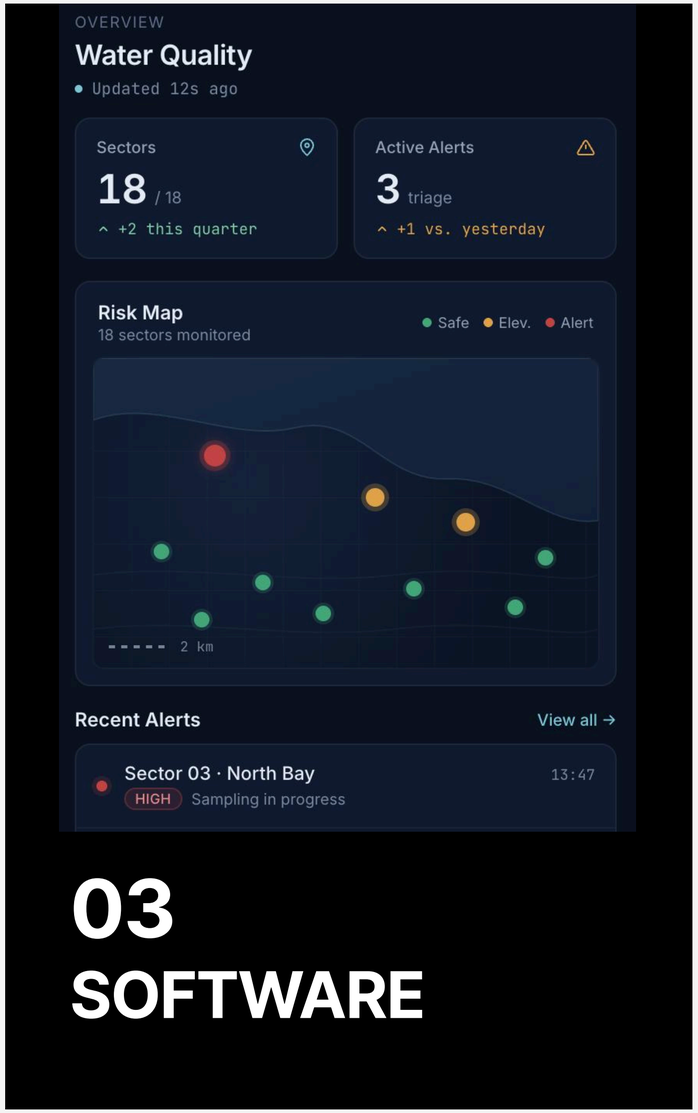
STORYBOARD



01
SMART BUOY



02
ML PREDICTION



03
SOFTWARE

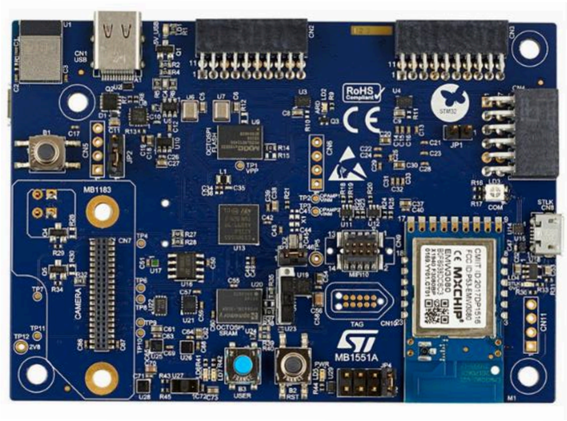
PREDICT EARLY
VERIFY FAST


LEVANTE CLOUD
REAL-TIME INTELLIGENCE
00:00
06:00
12:00
18:00
24.00
92
HEALTH
WATER QUALITY
Turbidity
pH
Conductivity
Dissolved Oxygen
OCEAN DYNAMICS
00:00
06:00
12:00
18:00
24.00
LEVANTE
EDGE AI
PRECIPITATION
Rain Gauge
WEATHER
Ultrasonic Mast
POSITION
GNSS Antenna
TELEMETRY
Wireless Link
WATER QUALITY
Optical Turbidity
OCEAN DYNAMICS
Current Velocity
TEMPERATURE
High-Precision Probe
ENVIRONMENTAL
Sensors
pH, Conductivity,

HARDWARE

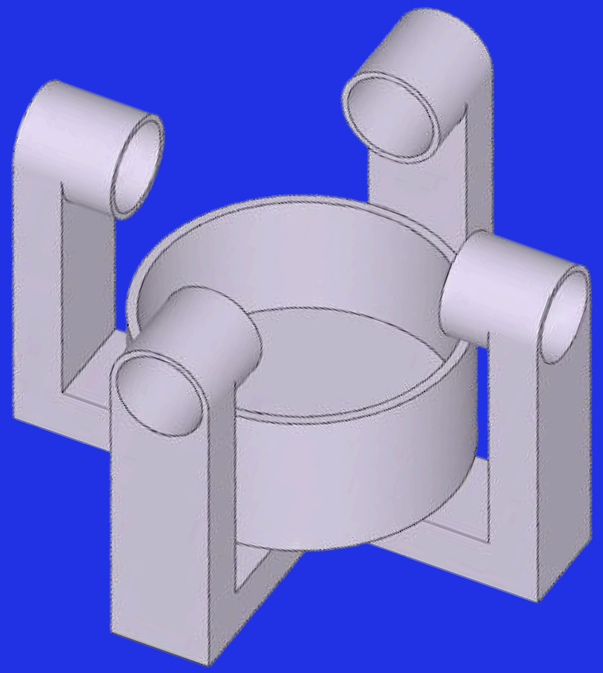






B-U585I-IOT02A

MODELING



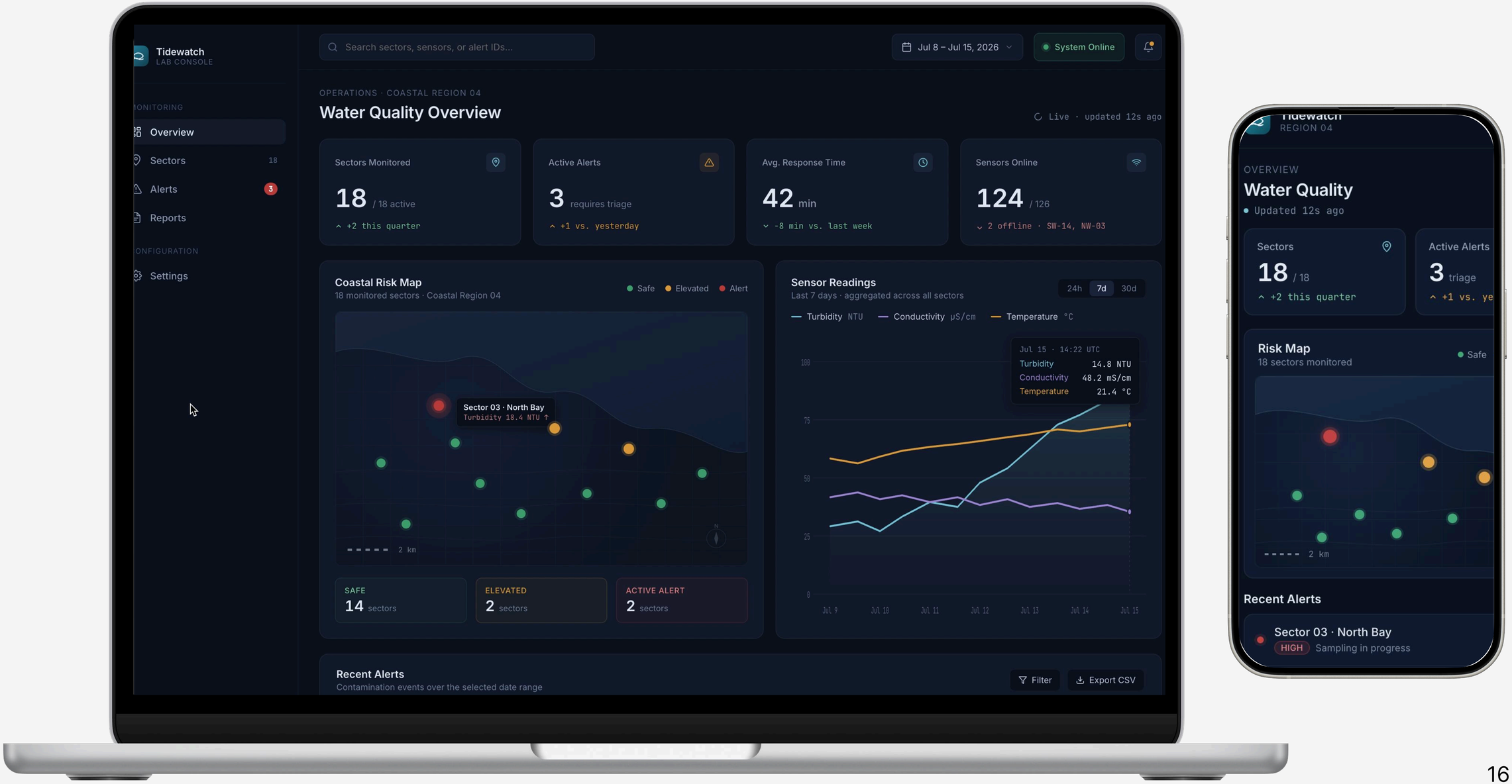
SOFTWARE

STM32 CubeIDE

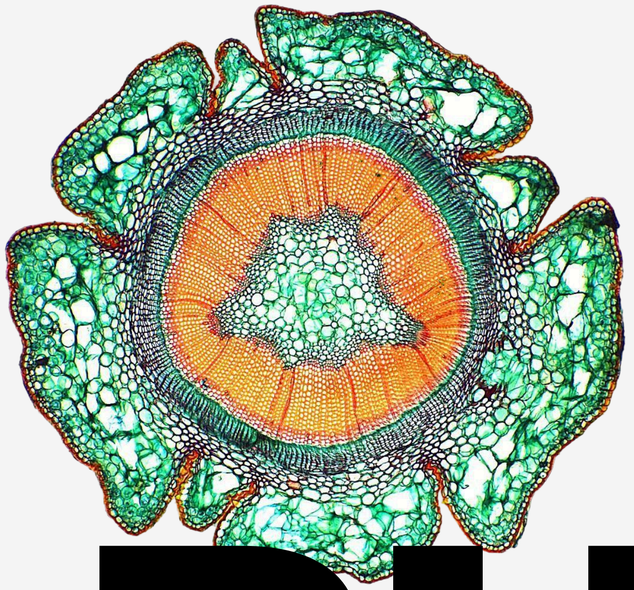
NANOEDGE AI STUDIO

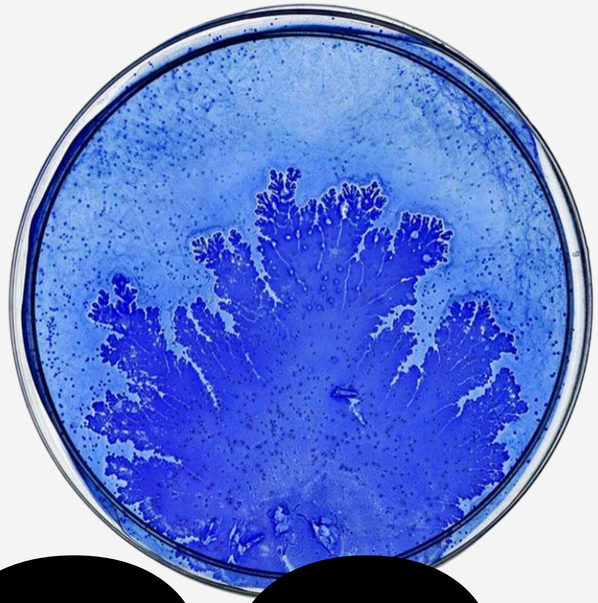


CapCut

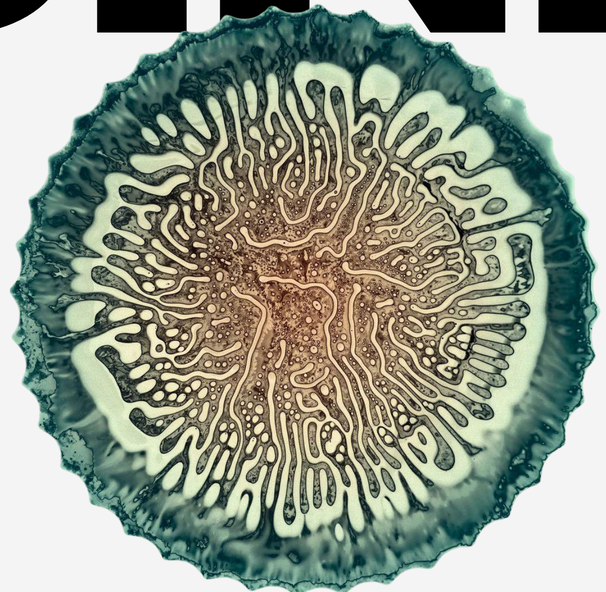

Tidewatch
LAB CONSOLE
Search sectors, sensors, or alert IDs...
Jul 8 – Jul 15, 2026
System Online
Overview
Sectors 18
Alerts 3
Reports
Settings
OPERATIONS · COASTAL REGION 04
Water Quality Overview
Live · updated 12s ago
Sectors Monitored
18 / 18 active
+2 this quarter
Active Alerts
3 requires triage
+1 vs. yesterday
Avg. Response Time
42 min
-8 min vs. last week
Sensors Online
124 / 126
2 offline · SW-14, NW-03
Coastal Risk Map
18 monitored sectors · Coastal Region 04
Safe
Elevated
Alert
Sector 03 · North Bay
Turbidity 18.4 NTU ↑
2 km
SAFE
14 sectors
ELEVATED
2 sectors
ACTIVE ALERT
2 sectors
Sensor Readings
Last 7 days · aggregated across all sectors
24h
7d
30d
Turbidity NTU
Conductivity µS/cm
Temperature °C
Jul 15 · 14:22 UTC
Turbidity 14.8 NTU
Conductivity 48.2 mS/cm
Temperature 21.4 °C
100
75
50
25
0
Jul 9
Jul 10
Jul 11
Jul 12
Jul 13
Jul 14
Jul 15
Recent Alerts
Contamination events over the selected date range
Filter
Export CSV

REGION 04
OVERVIEW
Water Quality
Updated 12s ago
Sectors
18 / 18
+2 this quarter
Active Alerts
3 triage
+1 vs. ye
Risk Map
18 sectors monitored
Safe
2 km
Recent Alerts
Sector 03 · North Bay
HIGH
Sampling in progress





BUSINESS

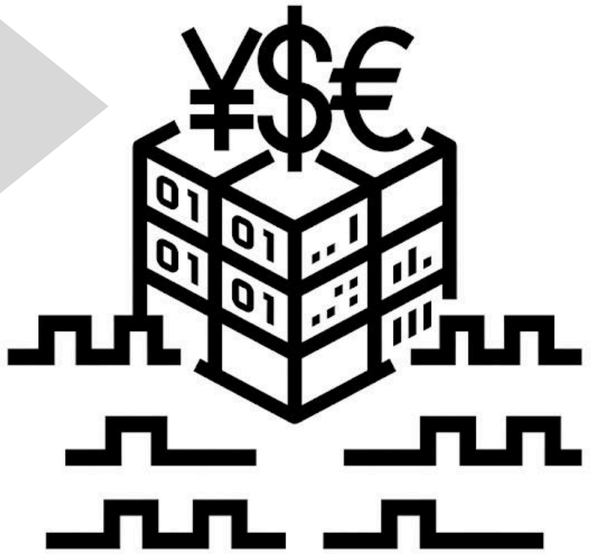




WE
Buoy + Analysis

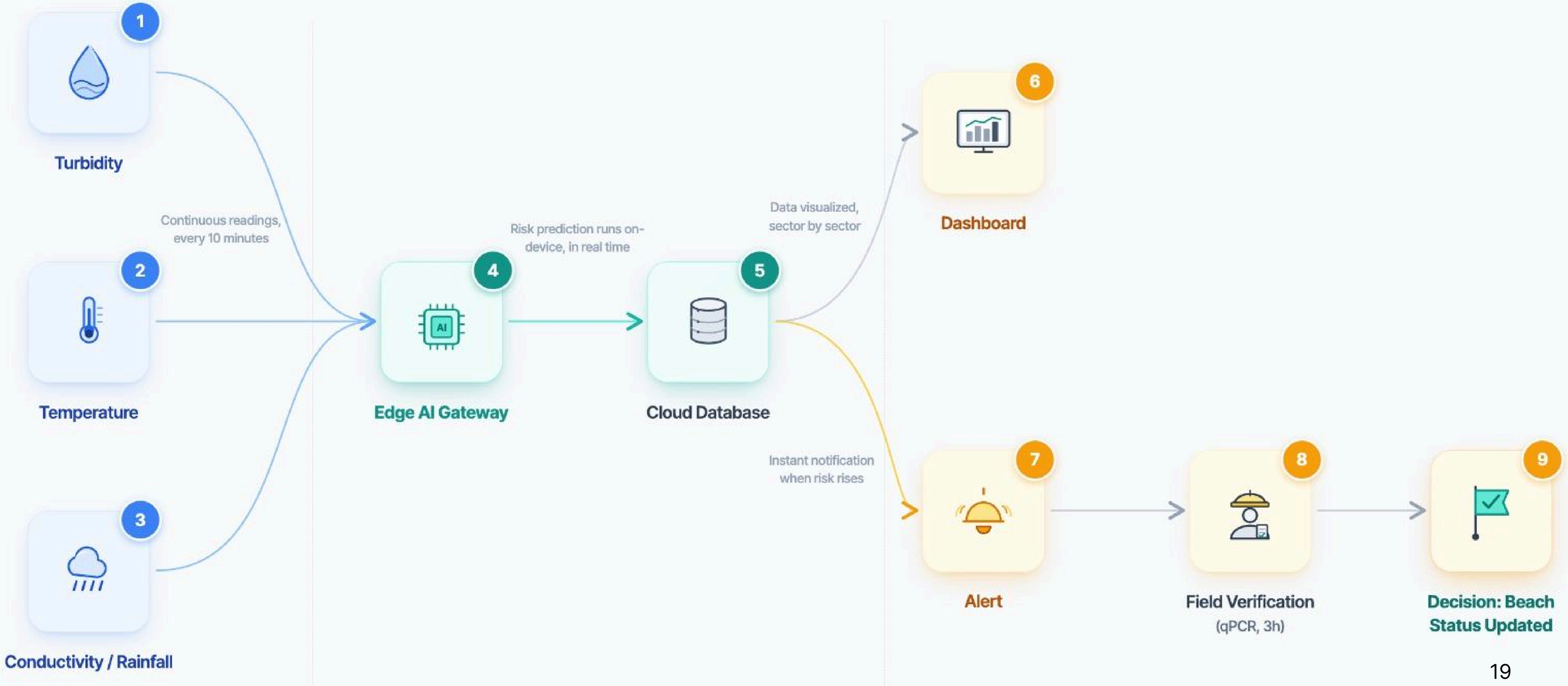


COSTUMER
Eurofins

REVENUE
Fee



Dynamic Architechture



Separable
Components

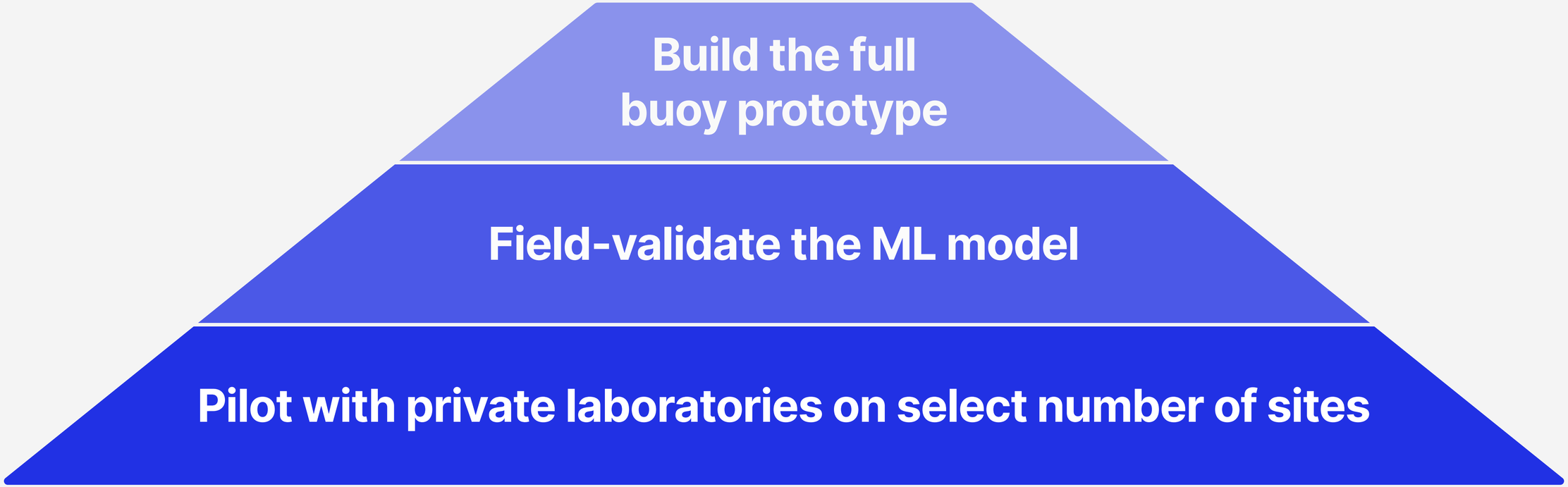
10-year
Chip Availability



Solar
Powered

Calibration
Data Transfer

WHAT'S NEXT


Build the full
buoy prototype
Field-validate the ML model
Pilot with private laboratories on select number of sites

24 Hours

50% OFF

THANKS

Group 3

Victor G. De Matteis
Giorgio Perakis
Raul Frandes
Carolina Mata
Riccardo Cirillo
Peng Ying

